

An empirical example: Four terminals and two areas

Daniel R. Miranda-Esquivel

2024 / 07 / 24

```
# Four taxa and two areas
```

```
## Libraries
```

```
library(blepd)
```

```
## Loading required package: ape
```

```
## Loading required package: picante
```

```
## Loading required package: vegan
```

```
## Loading required package: permute
```

```
##
```

```
## Attaching package: 'permute'
```

```
## The following object is masked from 'package:devtools':
```

```
##
```

```
##      check
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-6.1
```

```
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
```

```
##
```

```
## If you use the ggtree package suite in published research, please cite
```

```
## the appropriate paper(s):
```

```
##
```

```
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
```

```
## ggtree: an R package for visualization and annotation of phylogenetic
```

```
## trees with their covariates and other associated data. Methods in
```

```
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
```

```
##
```

```
## Guangchuang Yu, Tommy Tsan-Yuk Lam, Huachen Zhu, Yi Guan. Two methods
```

```
## for mapping and visualizing associated data on phylogeny using ggtree.
```

```
## Molecular Biology and Evolution. 2018, 35(12):3041-3043.
```

```
## doi:10.1093/molbev/msy194
```

```
##
```

```
## S Xu, Z Dai, P Guo, X Fu, S Liu, L Zhou, W Tang, T Feng, M Chen, L
```

```
## Zhan, T Wu, E Hu, Y Jiang, X Bo, G Yu. ggtreeExtra: Compact
```

```

## visualization of richly annotated phylogenetic data. Molecular Biology
## and Evolution. 2021, 38(9):4039-4042. doi: 10.1093/molbev/msab166

##
## Attaching package: 'ggtree'

## The following object is masked from 'package:nlme':
##
##      collapse

## The following object is masked from 'package:ape':
##
##      rotate

library(ggplot2)

## Data

# Call the 4 terminals example

data(package = "blepd")

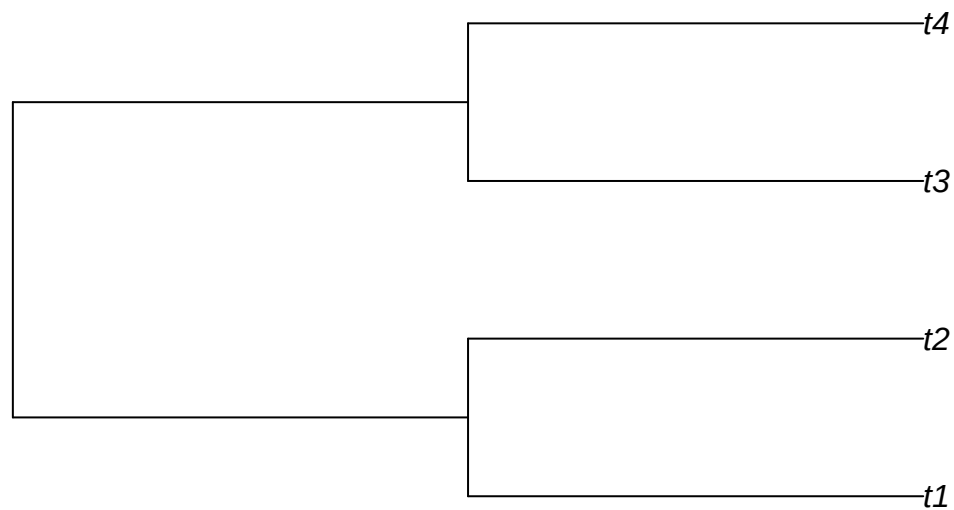
## trees

data(tree)

#str(tree)

plot(tree,use.edge.length =T)

```



```

initialTree <- tree

## distributions

data(distribution)

##str(distribution)

dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY

##   Terminal Area
## 1      t1    A1
## 2      t3    A1
## 3      t2    A2
## 4      t4    A2

## plotting

## the tree

```

```
plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals, all branches are == 1")

##
print(plotTree)
```

Four terminals, all branches are == 1

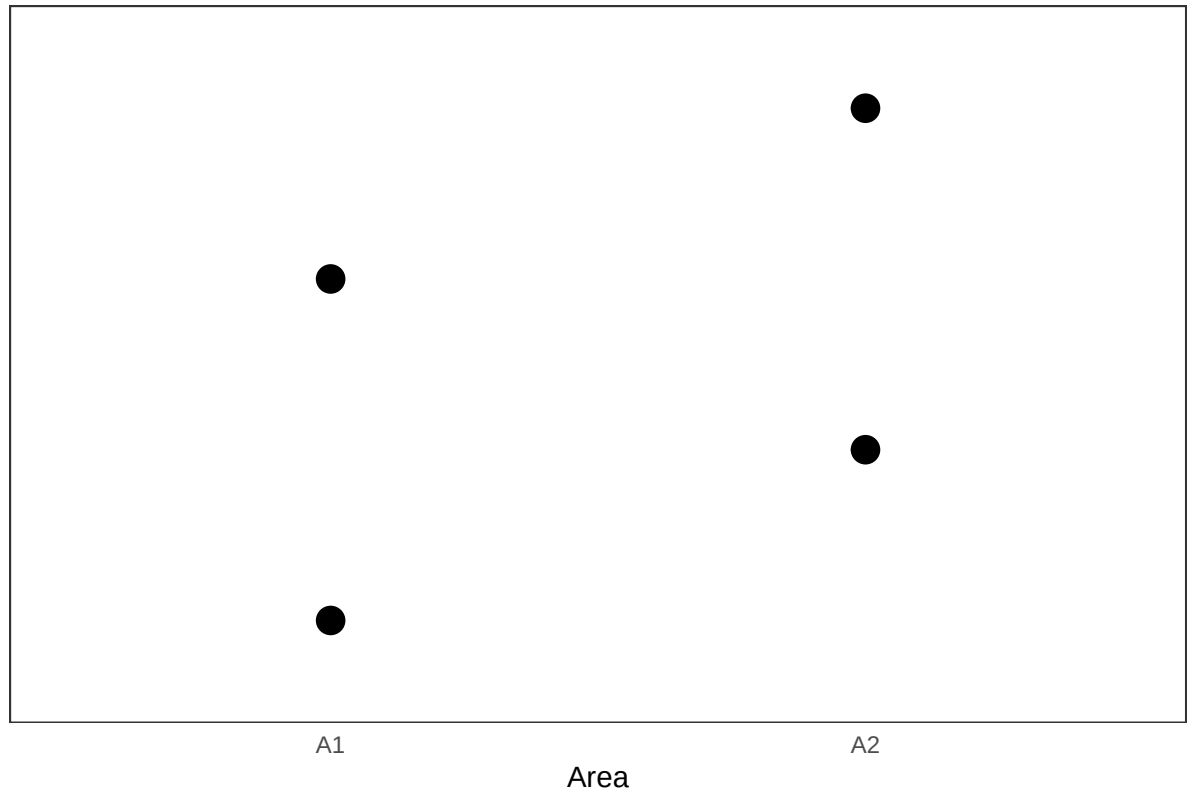


```
## the distribution

plotDistrib <- ggplot(data=distXY,
                     aes(x= Area, y= Terminal , size =150)) +
  geom_point() +
  theme_bw() +
  theme( axis.text.y = element_blank(),
        panel.grid.major = element_blank(), # Remove major gridlines
        panel.grid.minor = element_blank(), # Remove minor gridlines
        axis.line = element_line(colour = NA_character_), # Remove axis lines
        axis.ticks = element_blank(), # Remove axis ticks
        legend.position = "none"
  ) +
  labs(title = "Distributions",
       y = "",
```

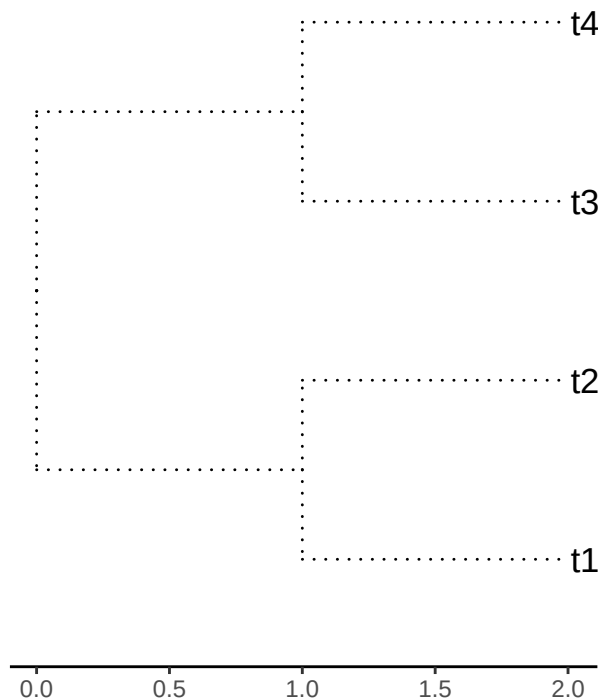
```
x = "Area")  
  
##  
print(plotDistrib)
```

Distributions

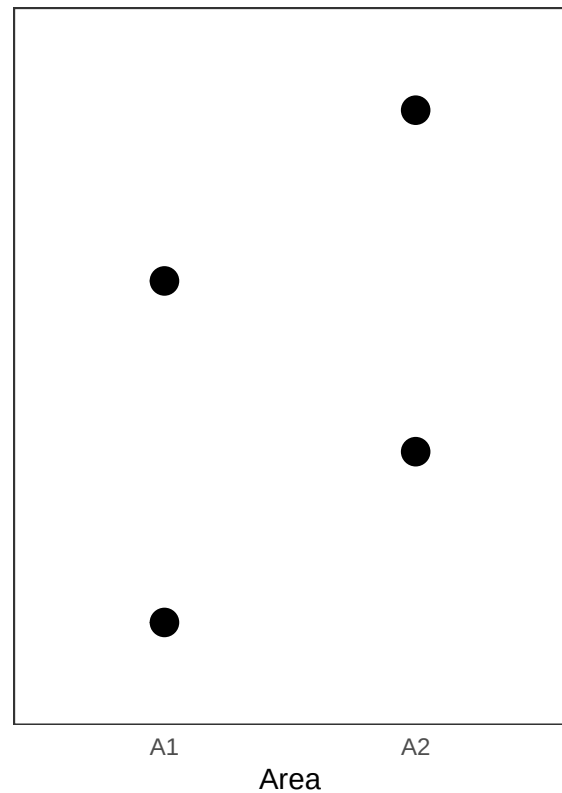


```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```

Four terminals, all branches are == 1



Distributions



```
#We check whether names in both objects: initialTree and dist4taxa, are the same.
```

```
##all(colnames(dist4taxa) == initialTree$tip.label)
```

```
all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```
#We report the branch length, and calculate the PD values.
```

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)
```

```
initialPD
```

```
## [1] 4 4
```

```
#As expected there is a tie between both areas.
```

```
#Now, we can see the impact of swapping the branch lengths; we will use the three  
#available models to swap: "simpleswap", "allswap", "uniform", and we will swap the  
#three classes of branches: "terminals", "internals", "all".
```

```
##?swapBL
```

```
for( modelo in c("simpleswap","allswap","uniform") ){  
  for( rama in c("terminals","internals","all") ){  
cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )  
    val <- swapBL(tree=initialTree,  
                  distribution = dist4taxa,  
                  model = modelo,  
                  branch = rama  
                  )  
    printBlepd(val)  
    cat( "\n\n" )  
  }  
}
```

```
##  
## Branchs swapped= terminals. model to test simpleswap reps 100  
##  
## BestInitial:A1A2  
## AreaSelected Freq Percent  
## 1 A1A2 100 100  
##  
##  
##  
## Branchs swapped= internals. model to test simpleswap reps 100  
##  
## BestInitial:A1A2  
## AreaSelected Freq Percent  
## 1 A1A2 100 100  
##  
##  
##  
## Branchs swapped= all. model to test simpleswap reps 100  
##  
## BestInitial:A1A2  
## AreaSelected Freq Percent  
## 1 A1A2 100 100  
##  
##  
##  
## Branchs swapped= terminals. model to test allswap reps 100  
##  
## BestInitial:A1A2  
## AreaSelected Freq Percent  
## 1 A1A2 100 100  
##  
##  
##
```

```

##
##   Branchs swapped=   internals.   model to test allswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   all.      model to test allswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   all.      model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100

```

#As expected, as all braches are EQUAL nothing happens

#Now, let us see if the results could be assigned to the longest branches (or not).

*## we could use a single command (*evalBranch*)*

##?evalBranch

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",

```



```

approach="all")

## Loading required package: maps
##
## Attaching package: 'phytools'
## The following object is masked from 'package:vegan':
##
##      scores
## let's see the structure of the output

##str(AllTerminals)

## Now we can print a far more friendly output

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Approach %Delta
## 1 [t1]|1          A1A2          A2   lower -1.024
## 2 [t2]|2          A1A2          A1   lower -1.024
## 3 [t3]|3          A1A2          A2   lower -1.024
## 4 [t4]|4          A1A2          A1   lower -1.024
## 5 [t1]|1          A1A2          A1   upper  2.072
## 6 [t2]|2          A1A2          A2   upper  2.072
## 7 [t3]|3          A1A2          A1   upper  2.072
## 8 [t4]|4          A1A2          A2   upper  2.072

## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##      node initialArea FinalArea Approach %Delta
## 1 [node number:6:t1/t2]|6          A1A2          A1A2   lower    0
## 2 [node number:7:t3/t4]|7          A1A2          A1A2   lower    0
## 3 [node number:6:t1/t2]|6          A1A2          A1A2   upper    0
## 4 [node number:7:t3/t4]|7          A1A2          A1A2   upper    0

##now with a longer branch

initialTree <- tree

initialTree$edge.length[1] <- 5

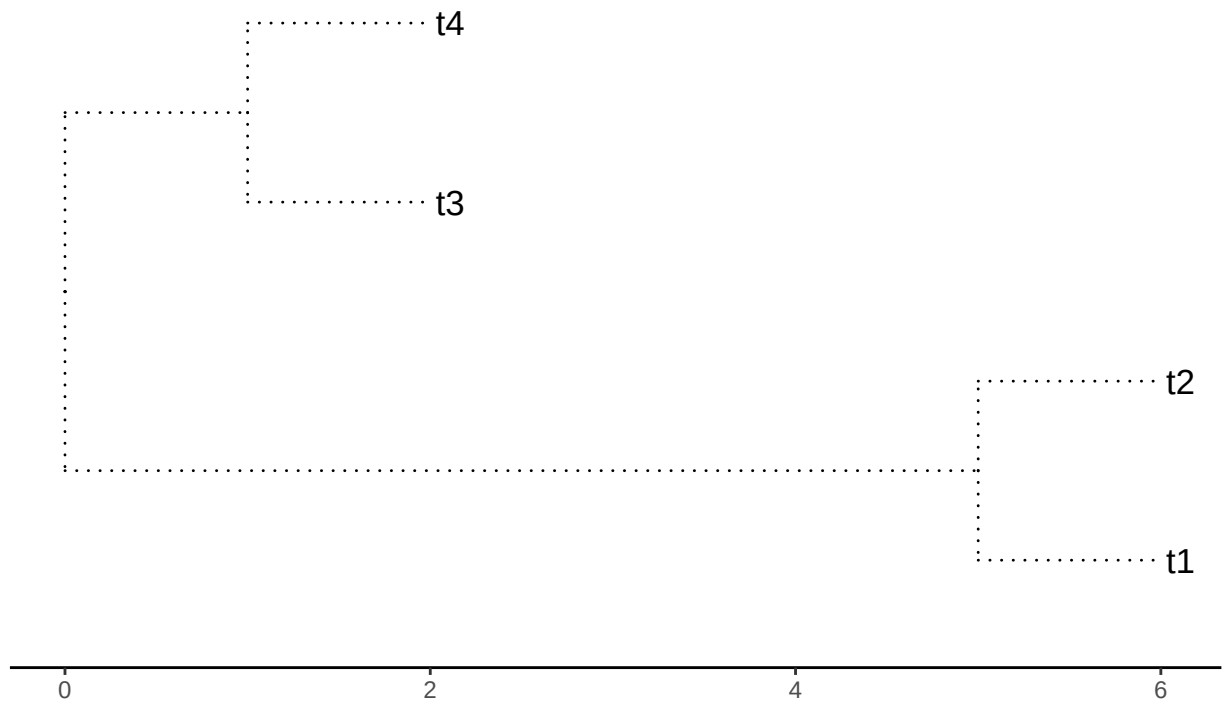
plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +

```

```
ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
". 2=",initialTree$edge.length[2],".",sep=""))
```

```
##
print(plotTree)
```

Four terminals. Branches 1=5. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printBlepd(val)
    cat( "\n\n" )
  }
}
```

```
##
```

```

## Branchs swapped= terminals. model to test simpleswap reps 100
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1      A1A2 100      100
##
##
##
## Branchs swapped= internals. model to test simpleswap reps 100
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1      A1A2 100      100
##
##
##
## Branchs swapped= all. model to test simpleswap reps 100
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1      A1      9      9
## 2      A1A2   83     83
## 3      A2      8      8
##
##
##
## Branchs swapped= terminals. model to test allswap reps 100
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1      A1A2 100      100
##
##
##
## Branchs swapped= internals. model to test allswap reps 100
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1      A1A2 100      100
##
##
##
## Branchs swapped= all. model to test allswap reps 100
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1      A1     33     33
## 2      A1A2   35     35
## 3      A2     32     32
##
##
##
## Branchs swapped= terminals. model to test uniform reps 100
##

```

```

## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=  internals.  model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=  all.      model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    52      52
## 2           A2    48      48

```

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

```

```

printEvalBranch(AllTerminals, compact=FALSE)

```

```

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1           A1A2          A2  lower -1.024
## 2 [t2]|2           A1A2          A1  lower -1.024
## 3 [t3]|3           A1A2          A2  lower -1.024
## 4 [t4]|4           A1A2          A1  lower -1.024
## 5 [t1]|1           A1A2          A1  upper  2.066
## 6 [t2]|2           A1A2          A2  upper  2.066
## 7 [t3]|3           A1A2          A1  upper  2.066
## 8 [t4]|4           A1A2          A2  upper  2.066

```

Now the internals

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

```

```

printEvalBranch(AllInternals, compact=FALSE)

```

```

##      node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6           A1A2          A1A2  lower    0
## 2 [node number:7:t3/t4]|7           A1A2          A1A2  lower    0
## 3 [node number:6:t1/t2]|6           A1A2          A1A2  upper    0
## 4 [node number:7:t3/t4]|7           A1A2          A1A2  upper    0

```

##now with a longer branch

```

initialTree <- tree

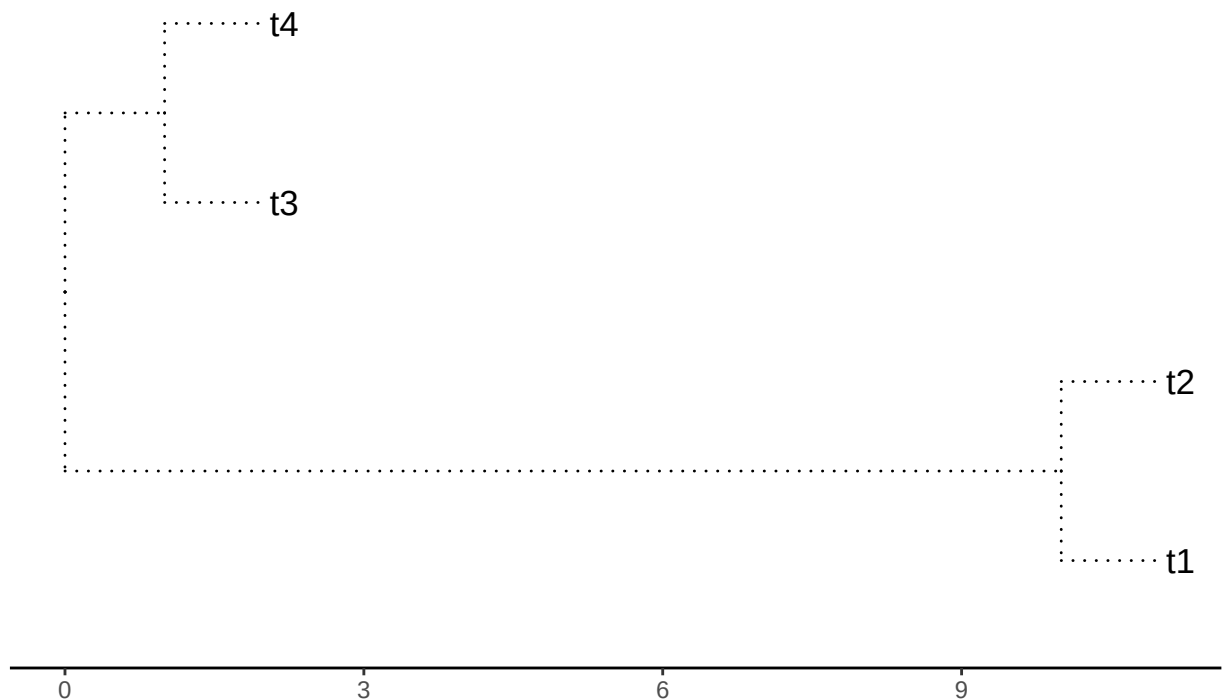
initialTree$edge.length[1] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
    ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=10. 2=1.



```

for( modelo in c("simpleswap","allswap","uniform") ){

  for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama

```

```

    )
    printBlepd(val)

    cat( "\n\n" )

}

}

```

```

##
##   Branchs swapped=   terminals.   model to test simpleswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   internals.   model to test simpleswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   all.        model to test simpleswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1   12       12
## 2           A1A2 75       75
## 3           A2   13       13
##
##
##
##   Branchs swapped=   terminals.   model to test allswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   internals.   model to test allswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   all.        model to test allswap reps 100
##
##

```

```

## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1   42      42
## 2          A1A2   36      36
## 3           A2   22      22
##
##
##
##   Branchs swapped=  terminals.  model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2  100      100
##
##
##
##   Branchs swapped=  internals.  model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2  100      100
##
##
##
##   Branchs swapped=  all.      model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1   57      57
## 2           A2   43      43

```

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

```

```

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1          A1A2          A2  lower -1.024
## 2 [t2]|2          A1A2          A1  lower -1.024
## 3 [t3]|3          A1A2          A2  lower -1.024
## 4 [t4]|4          A1A2          A1  lower -1.024
## 5 [t1]|1          A1A2          A1  upper  2.032
## 6 [t2]|2          A1A2          A2  upper  2.032
## 7 [t3]|3          A1A2          A1  upper  2.032
## 8 [t4]|4          A1A2          A2  upper  2.032

```

Now the internals

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

```

```

printEvalBranch(AllInternals, compact=FALSE)

##
##      node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6      A1A2      A1A2   lower      0
## 2 [node number:7:t3/t4] |7      A1A2      A1A2   lower      0
## 3 [node number:6:t1/t2] |6      A1A2      A1A2   upper      0
## 4 [node number:7:t3/t4] |7      A1A2      A1A2   upper      0
#now with a longer branch

initialTree <- tree

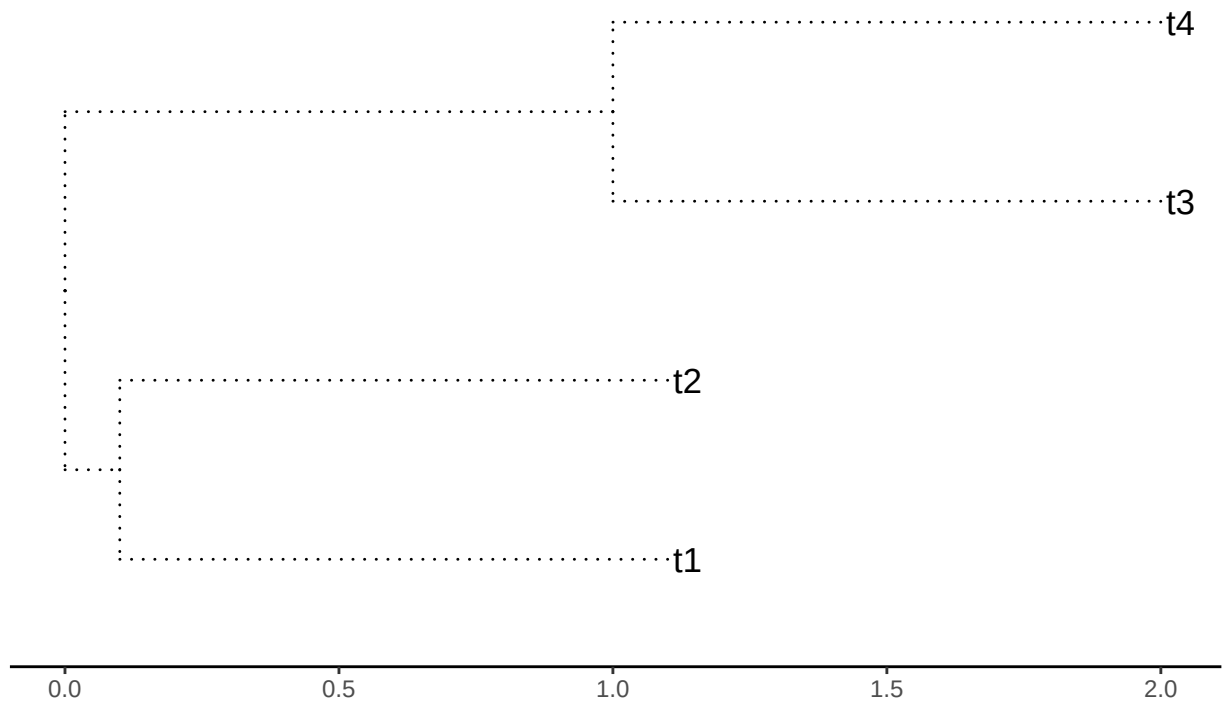
initialTree$edge.length[1] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```


Four terminals. Branches 1=0.1. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                  )
    printBlepd(val)

    cat( "\n\n" )

  }
}
```

```
##
##   Branchs swapped=   terminals.   model to test simpleswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
```

```

##
##   Branchs swapped=   internals.   model to test simpleswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   all.      model to test simpleswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    10       10
## 2           A1A2  76       76
## 3           A2   14       14
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100       100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100       100
##
##
##
##   Branchs swapped=   all.      model to test allswap reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    30       30
## 2           A1A2  42       42
## 3           A2   28       28
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100       100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100

```

```

##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    44      44
## 2           A2    56      56

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1           A1A2          A2   lower -1.024
## 2 [t2]|2           A1A2          A1   lower -1.024
## 3 [t3]|3           A1A2          A2   lower -1.024
## 4 [t4]|4           A1A2          A1   lower -1.024
## 5 [t1]|1           A1A2          A1   upper  2.061
## 6 [t2]|2           A1A2          A2   upper  2.061
## 7 [t3]|3           A1A2          A1   upper  2.061
## 8 [t4]|4           A1A2          A2   upper  2.061

## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6           A1A2          A1A2   lower    0
## 2 [node number:7:t3/t4]|7           A1A2          A1A2   lower    0
## 3 [node number:6:t1/t2]|6           A1A2          A1A2   upper    0
## 4 [node number:7:t3/t4]|7           A1A2          A1A2   upper    0

#now with a longer branch in brach 2

#now with a longer branch

initialTree <- tree

initialTree$edge.length[2] <- 5

```

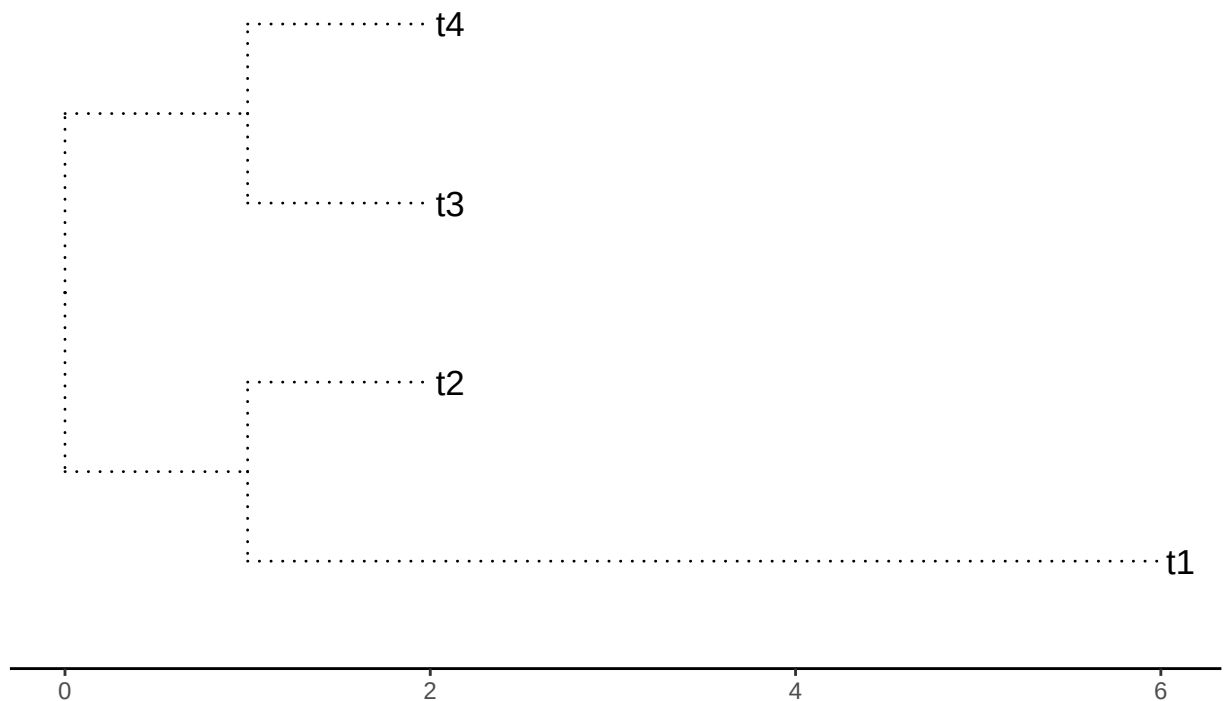
```

plotTree <- ggtree(initialTree, ladderize=TRUE,
                  color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=5.



```

for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printBlepd(val)

    cat( "\n\n" )
  }
}

```

```

    }
}

```

```

##
##   Branchs swapped=   terminals.   model to test simpleswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   68      68
## 2           A2   32      32
##
##
##
##   Branchs swapped=   internals.   model to test simpleswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100     100
##
##
##
##   Branchs swapped=   all.         model to test simpleswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   75      75
## 2          A1A2   11      11
## 3           A2   14      14
##
##
##
##   Branchs swapped=   terminals.   model to test allswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   42      42
## 2           A2   58      58
##
##
##
##   Branchs swapped=   internals.   model to test allswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100     100
##
##
##
##   Branchs swapped=   all.         model to test allswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   33      33

```

```

## 2      A1A2  33    33
## 3      A2   34    34
##
##
##
##   Branchs swapped=   terminals.   model to test uniform reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1      A1    53     53
## 2      A2    47     47
##
##
##
##   Branchs swapped=   internals.   model to test uniform reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1      A1   100    100
##
##
##
##   Branchs swapped=   all.       model to test uniform reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1      A1    50     50
## 2      A2    50     50

```

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

```

```

printEvalBranch(AllTerminals, compact=FALSE)

```

```

##      node initialArea FinalArea Aproach  %Delta
## 1 [t1]|1      A1      A2   lower -80.193
## 2 [t2]|2      A1      A1   lower      0
## 3 [t3]|3      A1      A1   lower      0
## 4 [t4]|4      A1      A1   lower      0
## 5 [t1]|1      A1      A1   upper      0
## 6 [t2]|2      A1      A2   upper 405.005
## 7 [t3]|3      A1      A1   upper      0
## 8 [t4]|4      A1      A2   upper 405.005

```

```

## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

```

```

printEvalBranch(AllInternals, compact=FALSE)

```

```
##               node initialArea FinalArea Approach %Delta
## 1 [node number:6:t1/t2]|6          A1          A1   lower      0
## 2 [node number:7:t3/t4]|7          A1          A1   lower      0
## 3 [node number:6:t1/t2]|6          A1          A1   upper      0
## 4 [node number:7:t3/t4]|7          A1          A1   upper      0

#now with a longer branch

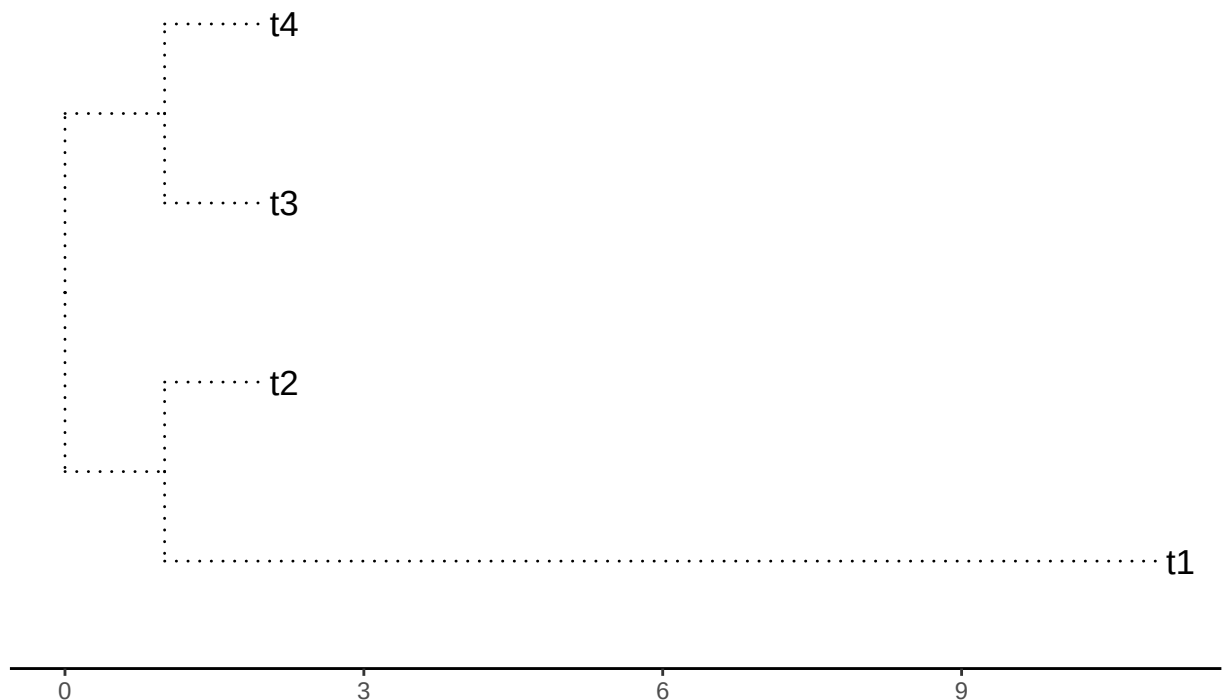
initialTree <- tree

initialTree$edge.length[2] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)
```

Four terminals. Branches 1=1. 2=10.



```
for( modelo in c("simpleswap","allswap","uniform") ){
```

```

    for( rama in c("terminals","internals","all") ){
cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                  )
    printBlepd(val)

    cat( "\n\n" )

    }

}

```

```

##
##   Branchs swapped=   terminals.   model to test simpleswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   65      65
## 2           A2   35      35
##
##
##
##   Branchs swapped=   internals.   model to test simpleswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100     100
##
##
##
##   Branchs swapped=   all.        model to test simpleswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   71      71
## 2          A1A2   16      16
## 3           A2   13      13
##
##
##
##   Branchs swapped=   terminals.   model to test allswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   47      47
## 2           A2   53      53
##
##
##
##   Branchs swapped=   internals.   model to test allswap reps 100

```



```

##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100      100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   30       30
## 2          A1A2   33       33
## 3           A2   37       37
##
##
##
##   Branchs swapped=  terminals.  model to test uniform reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   45       45
## 2           A2   55       55
##
##
##
##   Branchs swapped=  internals.  model to test uniform reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100      100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   53       53
## 2           A2   47       47

```

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

```

```

##      node initialArea FinalArea Aproach  %Delta
## 1 [t1]|1          A1          A2  lower -90.096
## 2 [t2]|2          A1          A1  lower      0
## 3 [t3]|3          A1          A1  lower      0
## 4 [t4]|4          A1          A1  lower      0
## 5 [t1]|1          A1          A1  upper      0
## 6 [t2]|2          A1          A2  upper 910.004

```

```
## 7 [t3]|3      A1      A1  upper      0
## 8 [t4]|4      A1      A2  upper 910.004
```

Now the internals

```
AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")
```

```
printEvalBranch(AllInternals, compact=FALSE)
```

```
##           node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6      A1      A1  lower      0
## 2 [node number:7:t3/t4]|7      A1      A1  lower      0
## 3 [node number:6:t1/t2]|6      A1      A1  upper      0
## 4 [node number:7:t3/t4]|7      A1      A1  upper      0
```

##now with a longer branch

```
initialTree <- tree
```

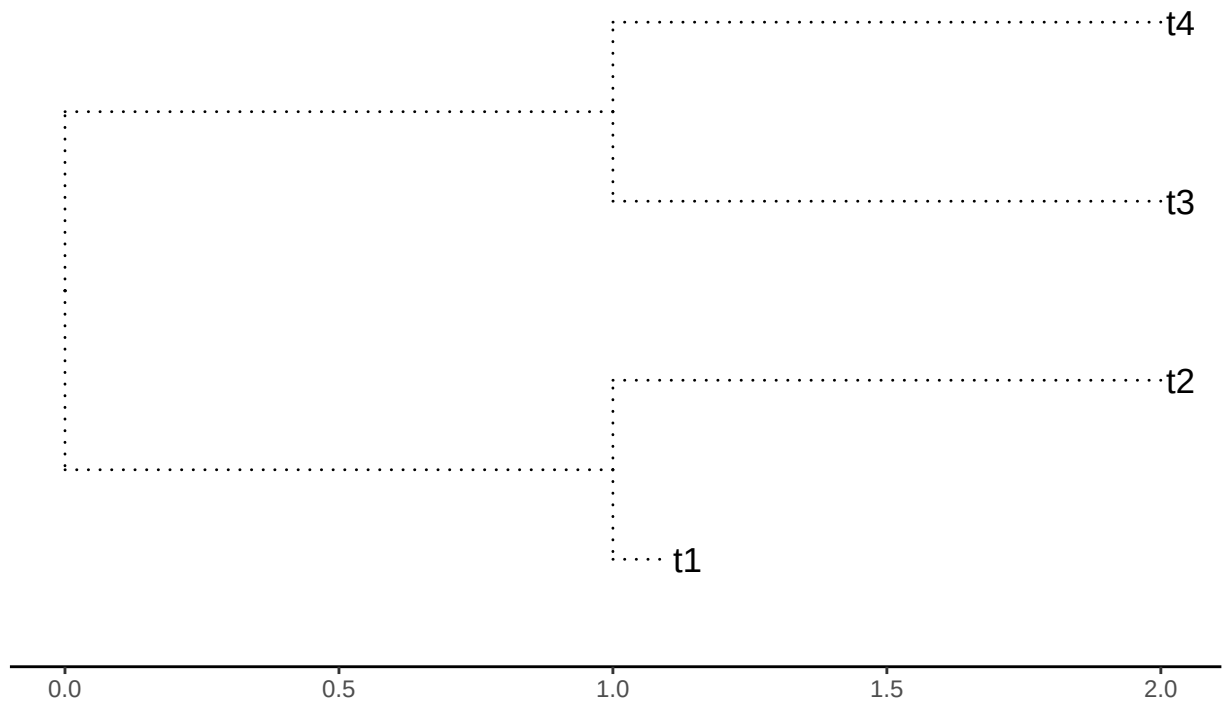
```
initialTree$edge.length[2] <- 0.1
```

```
plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))
```

##

```
print(plotTree)
```

Four terminals. Branches 1=1. 2=0.1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                  )
    printBlepd(val)
    cat( "\n\n" )
  }
}
```

```
##
##   Branchs swapped=   terminals.   model to test simpleswap reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   32      32
## 2           A2   68      68
##
```

```

##
##
##   Branchs swapped=   internals.   model to test simpleswap reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A2  100      100
##
##
##
##   Branchs swapped=   all.       model to test simpleswap reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   13       13
## 2          A1A2   18       18
## 3           A2   69       69
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   44       44
## 2           A2   56       56
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A2  100      100
##
##
##
##   Branchs swapped=   all.       model to test allswap reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   29       29
## 2          A1A2   32       32
## 3           A2   39       39
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   48       48
## 2           A2   52       52
##

```

```
##
##
##   Branchs swapped=   internals.   model to test uniform reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A2  100      100
##
##
##
##   Branchs swapped=   all.      model to test uniform reps 100
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   62       62
## 2           A2   38       38
```

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1           A2          A2  lower      0
## 2 [t2]|2           A2          A1  lower -90.115
## 3 [t3]|3           A2          A2  lower      0
## 4 [t4]|4           A2          A1  lower -90.115
## 5 [t1]|1           A2          A1  upper 910.412
## 6 [t2]|2           A2          A2  upper      0
## 7 [t3]|3           A2          A1  upper  91.905
## 8 [t4]|4           A2          A2  upper      0
```

Now the internals

```
AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)
```

```
##      node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6           A2          A2  lower      0
## 2 [node number:7:t3/t4]|7           A2          A2  lower      0
## 3 [node number:6:t1/t2]|6           A2          A2  upper      0
## 4 [node number:7:t3/t4]|7           A2          A2  upper      0
```